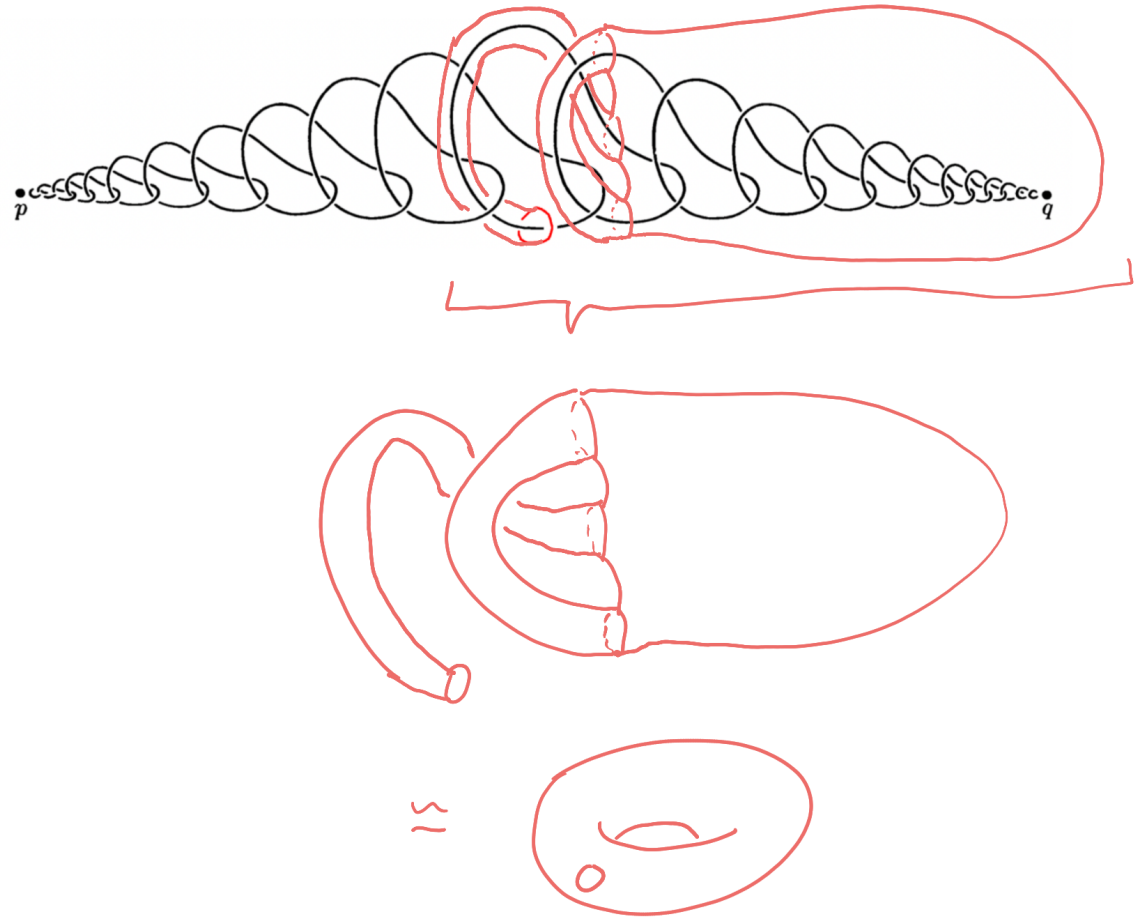
1.



2. I will prove that $\tilde{H}_k(BM((X_j, x_j)_1^m)) \cong \tilde{H}_k(BM((X_j, x_j)_1^{m-1})) \oplus \tilde{H}_k(X_m)$, in which case by induction we get $\tilde{H}_k(BM((X_j, x_j)_1^m)) \cong \bigoplus_{j=1}^m \tilde{H}_k(X_j)$. To prove this induction step, we use Mayer Vietoris, to specify the open sets denote the branches of the tree Y as $y_j \in I_j$, and let U_0 be a small open neighborhood of y_0, i.e. $U_0 = \bigcup_1^m [0, \epsilon)_j$ where $[0, \epsilon_j) \subset I_j$ and $\epsilon << 1$. for each (X_j, x_j) let $U_j = U_0 \cup I_j \cup X_j$, then clearly $U_j \simeq X_j$ by a deformation retract. We define $V = \bigcup_1^{m-1} U_j$, then $V \simeq BM((X_j, x_j)_1^{m-1})$ by a deformation retract, and denote $U = U_m$. Finally, we see that $U \cap V \simeq \{y_0\}$ by a deformation retract, so that applying Mayer Vietoris (on reduced homology) yields the exact sequence

$$\begin{tikzcd}
\cdots & \tilde{H}_{k-1}(U) \oplus \tilde{H}_{k-1}(V) \arrow[l] & \tilde{H}_{k-1}(U \cap V) \arrow[l] \arrow[dll, "\delta"'] \\
\tilde{H}_k(BM) & \tilde{H}_k(U) \oplus \tilde{H}_k(V) \arrow[l] & \tilde{H}_k(U \cap V) \arrow[l] \arrow[dll, "\delta"'] \\
\tilde{H}_{k+1}(BM) & \tilde{H}_{k+1}(U) \oplus \tilde{H}_{k+1}(V) \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

But since we are working with reduced homology, and and $U \cap V$ is connected, we get $\tilde{H}_k(U \cap V) =$

$\tilde{H}_k(\{y_0\}) = 0$ for all k, thus the above sequence simplifies to

$$\begin{tikzcd}
\cdots & \tilde{H}_{k-1}(U) \oplus \tilde{H}_{k-1}(V) \arrow[l] & 0 \arrow[l] \arrow[dll, "\delta"'] \\
\tilde{H}_k(BM) & \tilde{H}_k(U) \oplus \tilde{H}_k(V) \arrow[l] & 0 \arrow[l] \arrow[dll, "\delta"'] \\
\tilde{H}_{k+1}(BM) & \tilde{H}_{k+1}(U) \oplus \tilde{H}_{k+1}(V) \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

So that by exactness we have for all k,

$$\tilde{H}_k(BM((X_j, x_j)_1^m)) \cong \tilde{H}_k(U) \oplus \tilde{H}_k(V) \cong \tilde{H}_k(X_j) \oplus \tilde{H}_k(BM((X_j, x_j)_1^{m-1}))$$

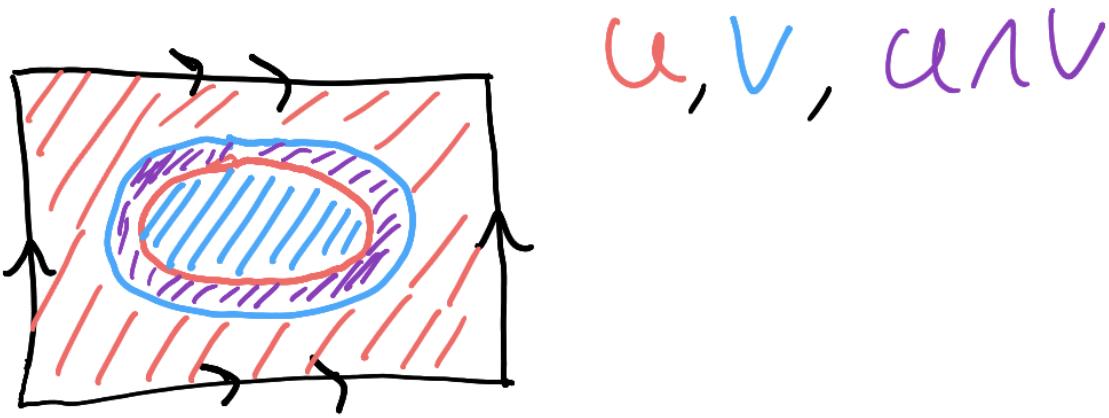
where we use homotopy invariance of homology for the second isomorphism. $\square$

3. Let $CX = X \times I/X \times \{1\}$ denote the mapping cone of X, then it is immediate that (X, CX) is a good pair (X includes along the zero coordinate) implying we get $\tilde{H}_k(CX, X) \cong \tilde{H}_k(\Sigma X)$, and that CX is contractible to $\{1\}$. Using this, we get that $\tilde{H}_k(CX) = 0$ for all k, and the following LES of the pair

$$\begin{tikzcd}
\cdots & 0 \arrow[l] & \tilde{H}_{k-1}(X) \arrow[l] \arrow[dll, "\delta"'] \\
\tilde{H}_k(\Sigma X) & 0 \arrow[l] & \tilde{H}_k(X) \arrow[l] \arrow[dll, "\delta"'] \\
\tilde{H}_{k+1}(\Sigma X) & 0 \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

Exactness gives us that $\tilde{H}_{k+1}(\Sigma X) \cong \tilde{H}_k(X)$, and $\tilde{H}_0(\Sigma X) = 0$. $\square$

4. (a) We use the following open sets and Mayer Vietoris



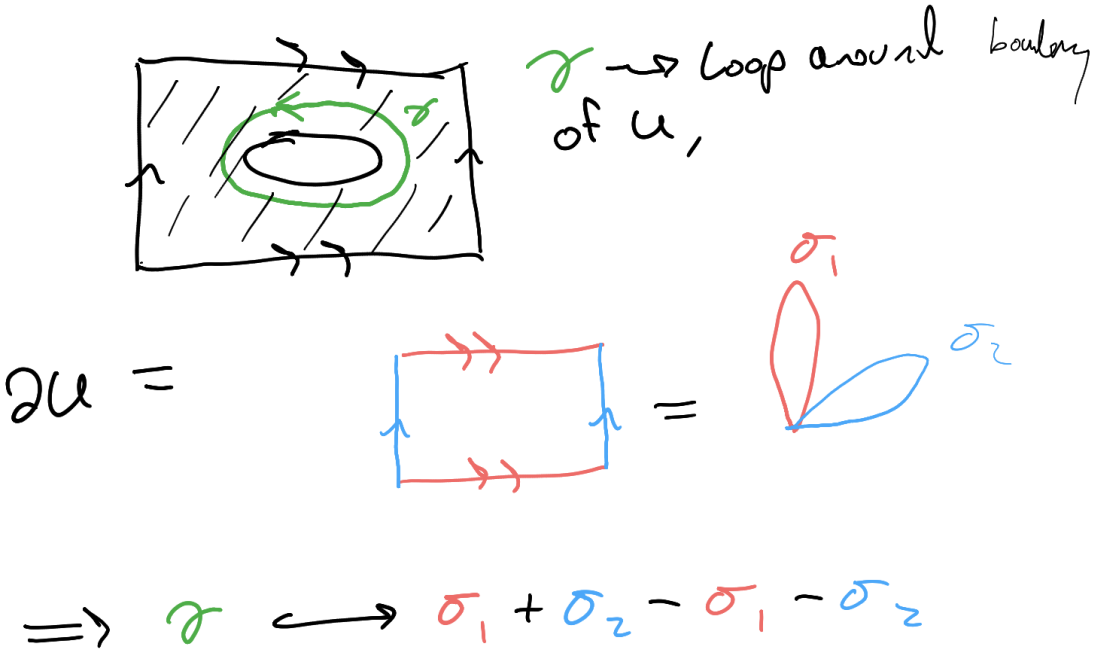
Then $U \simeq S^1 \vee S^1, V \simeq \{\text{pt.}\}$, and $U \cap V \simeq S^1$, thus $\tilde{H}_k(U) \cong \tilde{H}_k(S^1 \vee S^1), 0 = \tilde{H}_k(V)$ for all k, and $\tilde{H}_k(U \cap V) \cong \tilde{H}_k(S^1)$ for all k. We first compute $\tilde{H}_k(S^1 \vee S^1)$ also using Mayer Vietoris, for choosing U, V to be contractible neighborhoods of either copy of S^1, so that $U \cap V \simeq \{\text{pt}\}$ is a contractible neighborhood of the wedge point. Whence $\tilde{H}_k(U \cap V) = 0$ for all k, and $\tilde{H}_k(U) \cong \tilde{H}_k(S^1) \cong \tilde{H}_k(V)$ for all k. This gives the following long exact sequence using Mayer Vietoris:

$$\begin{tikzcd}
\cdots & \tilde{H}_{k-1}(S^1) \oplus \tilde{H}_{k-1}(S^1) \arrow[l] & 0 \arrow[l] \arrow[dll, "\delta"] \\
\tilde{H}_k(S^1 \vee S^1) & \tilde{H}_k(S^1) \oplus \tilde{H}_k(S^1) \arrow[l] & 0 \arrow[l] \arrow[dll, "\delta"] \\
\tilde{H}_{k+1}(S^1 \vee S^1) & \tilde{H}_{k+1}(S^1) \oplus \tilde{H}_{k+1}(S^1) \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

So that

$$\tilde{H}_k(S^1 \vee S^1) \cong \tilde{H}_k(S^1) \oplus \tilde{H}_k(S^1) \cong \begin{cases} \mathbb{Z} \oplus \mathbb{Z} & k = 1 \\ 0 & \text{else} \end{cases}$$

The last key ingredient we need is to compute $\iota_* : \tilde{H}_1(U \cap V) \to \tilde{H}_1(U) \oplus \tilde{H}_1(V)$. To do so, we use that $\sigma : \Delta^1 = I \to S^1$ via $x \mapsto e^{2\pi i x}$ has $[\sigma] = 1$. And moreover, by the above computation for $S^1 \vee S^1$, we may identify the generators for $\tilde{H}_1(S^1 \vee S^1)$ as this σ on either copy of S^1. Now we may use that the map $\sigma : I \to U \cap V$, which is the map with winding number one onto the copy of S^1 we are deformation retracting onto, satisfies that its image under the retract identifying U with $S^1 \vee S^1$ is $\sigma_1 + \sigma_2 - \sigma_1 - \sigma_2 = 0$, as in the following picture



This of course implies that $\iota_* : \tilde{H}_1(S^1) \to \tilde{H}_1(S^1 \vee S^1) \oplus 0$ is the zero map. Whence we get the

following Mayer-Vietoris sequence for T^2:

$$\begin{tikzcd}
\tilde{H}_0(T^2) & \tilde{H}_0(S^1 \vee S^1) \oplus 0 \arrow[l] & \tilde{H}_0(S^1) \arrow[l] \arrow[dll, "\delta"] \\
\tilde{H}_1(T^2) & \tilde{H}_1(S^1 \vee S^1) \oplus 0 \arrow[l] & \tilde{H}_1(S^1) \arrow[l, "0"'] \arrow[dll, "\delta"] \\
\tilde{H}_2(T^2) & \tilde{H}_2(S^1 \vee S^1) \oplus 0 \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

Cleaning things up further, we get all $\tilde{H}_k(S^1 \vee S^1) = 0 = \tilde{H}_k(S^1)$ for $k \geq 2$, so that $\tilde{H}_k(T^2) \cong 0$ for $k \geq 3$, and $\tilde{H}_0(T^2) = 0$ by connectedness. Writing everything up to isomorphism now gives

$$\begin{tikzcd}
0 & 0 \arrow[l] & 0 \arrow[l] \arrow[dll, "\delta"] \\
\tilde{H}_1(T^2) & \mathbb{Z} \oplus \mathbb{Z} \arrow[l] & \mathbb{Z} \arrow[l, "0"'] \arrow[dll, "\delta"] \\
\tilde{H}_2(T^2) & 0 \oplus 0 \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

So that since the map $\mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}$ is zero, we find that $\tilde{H}_2(T^2) \cong \mathbb{Z}$ and $\tilde{H}_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ by exactness. Combining everything we find that

$$\tilde{H}_k(T^2) \cong \begin{cases} \mathbb{Z} & k = 2 \\ \mathbb{Z} \oplus \mathbb{Z} & k = 1 \\ 0 & \text{else} \end{cases}$$

$\square$

(b) denote $M \simeq S^1$ to be the mobius strip, then let U_0 be a small neighborhood of $\{x_0\} \times S^1$ in T^2, which deformation retracts to $\{x_0\} \times S^1$ in T^2 and $U = U_0 \cup M \simeq M$, by simply retracting U_0 to $\{x_0\} \times S^1$. Let V_0 be a small neighborhood of ∂M, which retracts to the boundary, then let $V = V_0 \cup T^2 \simeq T^2$ by contracting V_0, moreover $U \cap V = U_0 \cup V_0 =\simeq \{x_0\} \times S^1$. This gives us the Mayer Vietoris sequence:

$$\begin{tikzcd}
\cdots & \tilde{H}_{k-1}(S^1) \oplus \tilde{H}_{k-1}(T^2) \arrow[l] & \tilde{H}_{k-1}(S^1) \arrow[l] \arrow[dll, "\delta"] \\
\tilde{H}_k(X) & \tilde{H}_k(S^1) \oplus \tilde{H}_k(T^2) \arrow[l] & \tilde{H}_k(S^1) \arrow[l] \arrow[dll, "\delta"] \\
\tilde{H}_{k+1}(X) & \tilde{H}_{k+1}(S^1) \oplus \tilde{H}_{k+1}(T^2) \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

since $\tilde{H}_k(S^1) = 0$, for $k \geq 2$, and $\tilde{H}_k(T^2) = 0$ for $k \geq 3$, we find that $\tilde{H}_k(X) = 0$ for $k \geq 3$. Morover, X is connected, so that $\tilde{H}_0(X) = 0$, we need only compute in degrees 1 and 2. By using the same standard generators for $\tilde{H}_1(S^1)$ as part (a) and the corresponding induced generators on $\tilde{H}_1(T^2)$, we have that $\iota_* : \tilde{H}_1(S^1) \to \tilde{H}_1(S^1) \oplus \tilde{H}_1(T^2)$ is identified with $\mathbb{Z} \to \mathbb{Z} \oplus (\mathbb{Z} \oplus \mathbb{Z}), 1 \mapsto (2, (0, 1))$ where the multiplication by 2 is induced by including to the boundary of the mobius strip being the winding twice around map under the homotopy identifying $M \simeq S^1$. This shows that ι_* is injective so we find that $\delta : \tilde{H}_2(X) \to \tilde{H}_1(S^1)$ is the zero map, and thus $\tilde{H}_2(X) \cong \tilde{H}_2(T^2) \cong \mathbb{Z}$ by exactness. It remains to compute in degree 1. Now we use that $j_* : \tilde{H}_1(S^1) \oplus \tilde{H}_1(T^2) \to \tilde{H}_1(X)$ satisfies $\ker j_* = \operatorname{Im} \iota_* = \mathbb{Z}(2, 0, 1)$, it follows that $j_* \mathbb{Z}^3 = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}$, but since $\tilde{H}_0(S^1) = 0$, we have $\delta : \tilde{H}_1(X) \to \tilde{H}_0(S^1)$ satisfies

$\ker\delta = \tilde{H}_1(X)$, so that by exactness $\tilde{H}_1(X) = \operatorname{Im} j_* \cong \mathbb{Z}^3/\mathbb{Z}(2,0,1)$, since we can take the following as a basis for $\mathbb{Z}^3$: $\{(2,0,1),(1,0,0),(0,1,0)\}$, we find that written in this basis $\tilde{H}_1(X) \cong \mathbb{Z}^3/\mathbb{Z} \oplus 0 \oplus 0 \cong \mathbb{Z}^2$. Everything taken together gives us

$$\tilde{H}_k(X) = \begin{cases} \mathbb{Z} & k = 2 \\ \mathbb{Z} \oplus \mathbb{Z} & k = 1 \\ 0 & \text{else} \end{cases}$$

□

5. Assume That $n \geq 1$, since the case for $n = 0$ is trivial for degrees of spheres, and points are non trivial on homology. **(a)** We use homework 5 problem 1, that a map of short exact sequences of chain complexes descends on homology to a map of the long exact sequences given by the snake lemma. This is proven in homework 5. We need a new naturality property, i.e. naturality of the good pair, this means that suppose we have (X, A) a good pair and $f : (X, A) \to (X, A)$ satisfying that the following commutes, for f' the induces map on the quotient

$$\begin{tikzcd} A \arrow[r, "\iota"] \arrow[d, "f"] & X \arrow[r, "\pi"] \arrow[d, "f"] & X/A \arrow[d, "f'"] \\ A \arrow[r, "\iota"] & X \arrow[r, "\pi"] & X/A \end{tikzcd}$$

Then given the following SES of short exact sequences, where once again $\hat{f}$ is the induces map

$$\begin{tikzcd} 0 \arrow[r] & C_*(A) \arrow[r, "\iota_*"] \arrow[d, "f_*"] & C_*(X) \arrow[r, "q"] \arrow[d, "f_*"] & C_*(X,A) \arrow[r] \arrow[d, "\hat{f}"] & 0 \\ 0 \arrow[r] & C_*(A) \arrow[r, "\iota_*"] & C_*(X) \arrow[r, "q"] & C_*(X,A) \arrow[r] & 0 \end{tikzcd}$$

The induced maps on homology from the functoriality of snake lemma satisfy $[\hat{f}] = [f']$ when $H_*(X, A)$ is identified with $H_*(X/A)$. To prove this, it suffices to show that $\pi : (X, A) \to (X/A, A/A)$ induces the isomorphism on homology, since then we get (since $n \neq 0$) $\pi_*[q] = \pi_*$, and thus $[\hat{f}] = \pi_*[q][f] = \pi_*[f] = [f']$. To see that $\pi : (X, A) \to (X/A, A/A)$ induces the isomorphism on homology, we use the following diagram, for U the open set containing A that retracts to A

$$\begin{tikzcd} H_n(X,A) \arrow[r] \arrow[d, "\pi_*"] & H_n(X,U) \arrow[d, "\pi_*"] & H_n(X \setminus A, U \setminus A) \arrow[l] \arrow[d, "\pi_*"] \\ H_n(X/A, A/A) \arrow[r] & H_n(X/A, U/A) & H_n(X/A \setminus A/A, U/A \setminus A/A) \arrow[l] \end{tikzcd}$$

The upper left horizontal is an isomorphism by the LES of the triple X, U, A (proven also in hw 5), since $(U, A) \simeq (A, A)$, similarly for the lower left. The upper right and lower right horizontal are ismorphisms by excision. The right vertical map is an isomorphism since π is the identity map on $X \setminus A$, the other two vertical maps follow from the 2-lemma. This gives the desired result. □

(b) Let $f : X \to X$, then we can define $Cf : CX \to CX$ as $\pi_{CX} \circ (f \times 1_I)$, and $\Sigma f : \Sigma X \to \Sigma X$ as $\pi_{\Sigma X} \circ (f \times 1)$ where $\pi_{CX}, \pi_{\Sigma X}$ are the quoients $X \times I \to CX$ and ΣX respectively. This gives us the following commutative diagram:

$$\begin{tikzcd} X \arrow[r, "\iota"] \arrow[d, "f"] & CX \arrow[r, "\pi"] \arrow[d, "Cf"] & \Sigma X \arrow[d, "\Sigma f"] \\ X \arrow[r, "\iota"] & CX \arrow[r, "\pi"] & \Sigma X \end{tikzcd}$$

Moreover, we get the following SES of chain complexes

$$\begin{tikzcd}
0 \arrow[r] & C_*(X) \arrow[r, "\iota_*"] \arrow[d, "f_*"] & C_*(CX) \arrow[r, "q"] \arrow[d, "Cf_*"] & C_*(CX, X) \arrow[r] \arrow[d, "\widehat{Cf}"] & 0 \\
0 \arrow[r] & C_*(X) \arrow[r, "\iota_*"] & C_*(CX) \arrow[r, "q"] & C_*(CX, X) \arrow[r] & 0
\end{tikzcd}$$

Using the solution to (a) gives us

$$\begin{tikzcd}
\cdots H_{n+1}(\Sigma X) \arrow[r] \arrow[d, "\Sigma f_*"] & H_n(X) \arrow[r] \arrow[d, "f_*"] & H_n(CX) \arrow[r] \arrow[d, "Cf_*"] & H_n(\Sigma X) \arrow[r] \arrow[d, "\Sigma f_*"] & H_{n-1}(X) \cdots \arrow[d, "f_*"] \\
\cdots H_{n+1}(\Sigma X) \arrow[r] & H_n(X) \arrow[r] & H_n(CX) \arrow[r] & H_n(\Sigma X) \arrow[r] & H_{n-1}(X) \cdots
\end{tikzcd}$$

Substituting $X = S^n$ and $\Sigma X \cong S^{n+1}$ we get the square

$$\begin{tikzcd}
H_{n+1}(S^{n+1}) \arrow[r, "\cong"] \arrow[d, "\Sigma f_*"] & H_n(S^n) \arrow[d, "f_*"] \\
H_{n+1}(S^{n+1}) \arrow[r, "\cong"] & H_n(S^n)
\end{tikzcd}$$

Then $\Sigma f_* \overset{\text{def.}}{=} \deg \Sigma f$ and $f_* \overset{\text{def.}}{=} \deg f$, commutativity of the diagram gives us that these degrees are in fact equal. $\square$

6. (a) Assume for contradiction $f(x) \neq \pm x$ for any x, embed S^n into $\mathbb{R}^{n+1}$, then define $\pi(x, y)$ to be the projection of y onto $T_x S^n = (\mathbb{R}x)^\perp \subset \mathbb{R}^{n+1}$, then $\pi(x, y) : \mathbb{R}^{n+1} \times \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ is continuous since it can be written as $(x, y) \mapsto y - \langle x, y \rangle x$, this implies the following is a composition of continuous functions and thus continuous

$$x \mapsto \pi(x, f(x) - x)$$

This is nonvanishing since $f(x) \neq \pm x$, so this map gives a nonvanishing continuous vectorfield on S^n, this contradicts the hairy ball theorem since n is even. $\square$

(b) Suppose $f : \mathbb{RP}^n \to \mathbb{RP}^n$ for n even, then $F : S^n \to \mathbb{RP}^n$ via $x \mapsto f \circ \rho(x)$ where ρ is the projection is continuous by definition of the quotient topology. Now since S^n is a covering space of $\mathbb{RP}^n$, the map F has a lift to $\widehat{F} : S^n \to S^n$ satisfying $\rho \circ \widehat{F} = F$, the construction of each of these maps gives commutativity of

$$\begin{tikzcd}
S^n \arrow[r, "\widehat{F}"] \arrow[d, "\rho"] & S^n \arrow[d, "\rho"] \\
\mathbb{RP}^n \arrow[r, "f"] & \mathbb{RP}^n
\end{tikzcd}$$

Now, by part (a) $\widehat{F}(x) = \pm x$ for some $x \in S^n$, so that $f([x]) = [\widehat{F}(x)] = [x]$. $\square$